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EXPERIMENT-1

Aim → To Determine the percentage composition of NaOH in the given mixture of NaOH and NaCl [8 g/L]

Apparatus Requirements → Burette, Pippette, conical flask, beakers

Chemical Requirements → 0.1 N HCl Sodium Hydroxide, sodium chloride, Phenolphthalein Indicator.

Principle →

For the titration of a solution of NaOH and NaCl, the other solution need is HCl solution. NaOH reacts with HCl solution whereas NaCl will remain as such

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

The mixture solution will be titrated with the acid solution of HCl and we can find out the normality and strength of NaOH. By dividing the strength of NaOH from the total known strength of the solution, we will find out the strength of NaCl and hence find the percentage composition.

Indicator → Phenolphthalein.

End point → Pink to Colourless.

Procedure →

1. Rinse the burette with the given N/10 HCl solution.
2. Take the HCl in the burette and note the initial reading.

Observations →

S No.	Burette Reading Initial	Burette Reading final	Volume of HCL [ml/L]	Concordant Reading
1	0	9.7	10ml	9.7
2	9.7	19.4	10ml	
3	19.4	29.1	10ml	
4				

Calculations →

By the law of Chemical equivalents,

$$N_1V_1 = N_2V_2$$

$$\frac{N_1V_1}{V_2} = N_2$$

$$\frac{0.1 \times 9.7}{10} = N_2$$

$$0.097 = N_2$$

Normality of given NaOH solution (N_2) = $N_2 \times 40 \Rightarrow 3.88$ g/L
Equivalent weight of NaOH = 3.88 g/L

Amount of NaCl in the given mixture = 4.12 g/L

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3. Pipette out 10 mL of the solution in the conical flask.
4. Add a drop of the indicator in the solution.
5. Add the acid solution from burette into the conical flask till the solution becomes ~~th~~ colourless.
6. Note the burette reading.
7. Repeat the experiment to get three concordent reading.

Result →

1. Normality of sodium Hydroxide solution: ~~3.88~~ 0.097
2. Strength of sodium hydroxide solution: 3.88 g/L
3. Percentage of NaOH: 52.5 %
4. Percentage of NaCl: 47.5 %

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Percentage Composition →

% composition of NaCl ⇒ $\frac{3.8}{8} \times 100$ ⇒ 47.5 %

% composition of NaOH ⇒ $\frac{4.12}{8} \times 100$ ⇒ 52.5 %

EXPERIMENT-2.

Aim:- To determine the alkalinity of a given water sample.

Apparatus Required:- Burette, pipette, conical flask and beakers

Chemicals Required:- N/10 HCl, Phenolphthalein and methyl orange.

Theory:- Alkalinity in water is due to presence of different ions in water. Determination of alkalinity in water is based on titration of water sample against a standard acid using selective indicators. The indicators used are phenolphthalein and methyl orange. The following reactions takes place :-

1 $OH^- + H^+ \rightarrow H_2O$
2 $CO_3^{2-} + H^+ \rightarrow HCO_3^-$
3. $HCO_3^- + H^+ \rightarrow H_2CO_3$

The volume of the acid used up to phenolphthalein end point ~~indicates~~ corresponds to the reaction [1] and [2] i.e complete neutralization of OH^- ions and CO_3^{2-} ions upto HCO_3^- stage. The volume of the acid used upto methyl orange end point corresponds to the reaction [1], [2], [3], ie complete neutralization of OH^-, CO_3^{2-} and HCO_3^- ions. Thus, from the volume of respective titration, the strengths of various ions can be determined. By measuring phenolphthalein ~~alkali~~ alkalinity and methyl orange alkalinity, it is possible to calculate the magnitude of various forms of alkalinity present in the water sample.

Eg. ① Alkalinity due to HCO_3^- only
② Alkalinity due to CO_3^{2-} only
③ Alkalinity due to CO_3^{2-} and HCO_3^-
④ Alkalinity due to CO_3^{2-} and OH^-
⑤ Alkalinity due to OH only

Observations →

Using phenolphthalein normality of the acid used = N/10

SNo.	Burette Readings		Volume of the titration used [Final - Initial Reading] [ml]
	Initial Reading	Final Reading	
1	0	6.1	6.1
2	6.1	12.2	6.1
3	12.2	18.4	6.2

Concordant Volume :- 6.2

Note: If no colour developes on addition of phenolphthalein to water sample, it means that phenolphthalein alkalinity is zero and hence do not titrate the sample for phenolphthalein alkalinity

<u>Case I</u>- When phenolphthalein alkalinity = 0, means OH^- and CO_3^{2-} both ions are absent. Whatever alkalinity is present due to HCO_3^- ions and can be detected using methyl orange as indicator.

<u>Case II</u>- when $P = M/2$, this means that only CO_3^{2-} ions are present. Neutralization reaches upto HCO_3^- ~~ions~~ stage using phenolphthalein as indicator. Some amount of acid will be further used to neutralize HCO_3^- to H_2O and CO_2. M or 2P will determine the strength of CO_3^{2-}.

<u>Case III</u> - When $P < M/2$, this means that in addition to CO_3^{2-} ions, HCO_3^- ions are also present.
Alkalinity due to $CO_3^{2-} = 2P$
Alkalinity due to $HCO_3^- = [M-2P]$

<u>Case IV</u>- when $P > M/2$, this means that in addition to CO_3^{2-} ions, HCO_3^- ions are also present.
Let $(OH^-) = x$, $(CO_3^{2-}) = 2y$
Then $P = x+y$ -----(i) and $M = x+2y$ ----(ii), so subtracting (i) from (ii), we get $(M-P) = y$ ----- (iii)
Putting the value of y from (ii) to (i), we get
$P = x + -P + M$, or
$(OH^-) = x = 2P - M$ ----(iv) and $(CO_3^{2-}) = 2y = 2(M-P)$

<u>Case V</u> - When $P = M$, this means that only (OH^-) ions are present. So $(OH^-) = P = M$

<u>Significance</u> - Highly alkaline waters are usually unpalatable and upper limits with respect to phenolphthalein alkalinity and total alkalinity is required to be specified. Alkaline water used in boilers for steam generation may lead to precipitation of sludges, deposition of scale and cause caustic

Using methyl orange, normality of the acid used = N/10

SNo.	Burette Reading		Volume of the titration used [Final - Initial Reading] [ml]
	Initial Reading	Final Reading	
1	0	23	23
2	23	46	23
3	46	68	22

Concordant volume = 23

Calculations :-

1. Phenolphthalein alkalinity in terms of $CaCO_3$ equivalent

$$N_1V_1 = N_2V_2$$

$$\frac{1}{10} \times 6.2 = N_2 \times 10$$

$$N_2 = \frac{6.2}{100} \Rightarrow \boxed{0.062}$$

Strength in terms of $CaCO_3$ equiv = $N_2 \times$ eq. wt of $CaCO_3$

$\Rightarrow 0.062 \times 60 \Rightarrow \boxed{3.1 g/L}$

$\Rightarrow \boxed{3100\ mg/L}$

embitterment, Knowledge of the kinds of alkalinity present in water and their magnitudes is important in calculating the amount of time $[Ca(OH)_2]$ and soda $[Na_2(CO_3)]$ needed for water softening. Use of different fertilisers is dictated by alkalinity of water.

Procedure -

① Pipette out 20 ml of water sample into a conical flask. Add 1-2 drops of phenolphthalein.
② Rinse and fill the burette with N/10 HCL.
③ Titrate the water sample in conical flask with N/10 HCL till the pink colour just disappears.
④ Note down the reading and repeat to get three concordent readings.
⑤ Again take 20 ml of the water sample in the flask and add 2-3 drops of methyl orange indicator to it.
⑥ Titrate it using N/10 HCL till just red colour is obtained.
⑦ Record the observation and repeat to get three concordent readings.

Result -

Phenolphthalein alkalinity = ~~2650~~ ppm
Methyl orange alkalinity = ppm
Alkalinity in individual terms = (i) CO_3^{2-} = 6200 ppm
(ii) HCO_3^- = 5300 ppm.

2. Methyl Orange Alkalinity in terms of $CaCO_3$ equivalent.

$$N_1V_1 = N_2V_2$$

$$\frac{1}{10} \times 23 = N_2 \times 10$$

$$N_2 = \frac{23}{100} \Rightarrow \boxed{0.23}$$

Strength in terms of $CaCO_3$ equivalent $= N_2 \times$ Eq. wt of $CaCO_3$

$$\Rightarrow 0.23 \times 50 \Rightarrow \boxed{11.5\,g/L}$$

$$\Rightarrow \boxed{11500\ mg/L}$$

P = 3100 ppm

M = 11500 ppm

M/2 = 5750 ppm

$P < \frac{1}{2}M$, so CO_3^{2-} ions & HCO_3^- ions are present

Alkalinity due to :-

$CO_3^{2-} \Rightarrow 2P \Rightarrow 2 \times 3100 \Rightarrow \boxed{6200\,ppm}$

$HCO_3^- \Rightarrow M - 2P \Rightarrow 11500 - 6200 \Rightarrow \boxed{5300\,ppm}$

EXPERIMENT-3

Aim:- To estimate the strength of calcium ions of the given solution using a standard solution of EDTA by complexometric solution.

Apparatus Required:- 100 ml standard flask, burette, 250 ml conical flask, 20 ml pipette and simple balance with weights.

Chemical Required:- Ethylene Diamine Tetra Acetic Acid (EDTA), Erichrome Black-T [EBT] Indicator, Ammonia buffer solution, hard water and distilled water.

Principle:- Hard water, which contains calcium and magnesium ions forms a wine red colour complex with the indicator, Erichrome Black-T, Ethylene di-amine Tetra Acetic acid [EDTA] forms a colourless stable complex with free metal ion like Ca.

Metal Indicator ⟶ Metal Indicator Complex.
[Wine Red Colour]

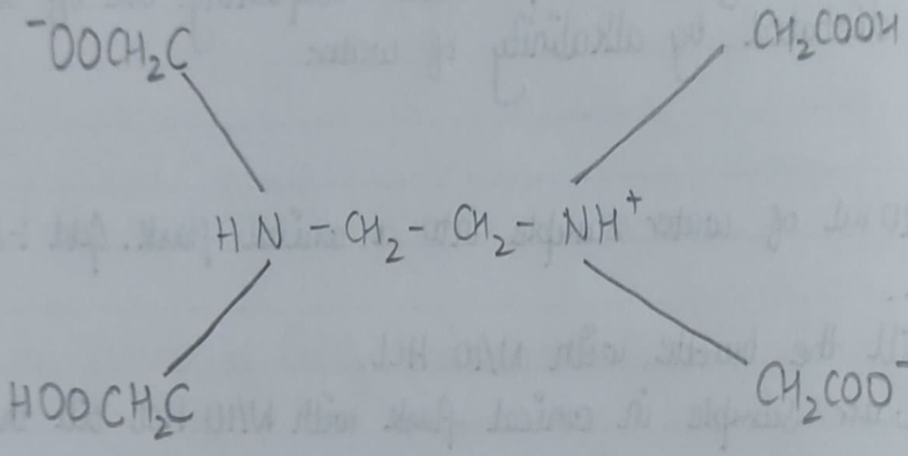
When EDTA is added from the burette, it extracts the metal ions from the metal ion indicator complex, thereby releasing the free indicator complex. The stability of metal ion indicator complex is less than that of a metal ion-EDTA complex.

And hence EDTA extracts metal ion from the ion indicator complex.

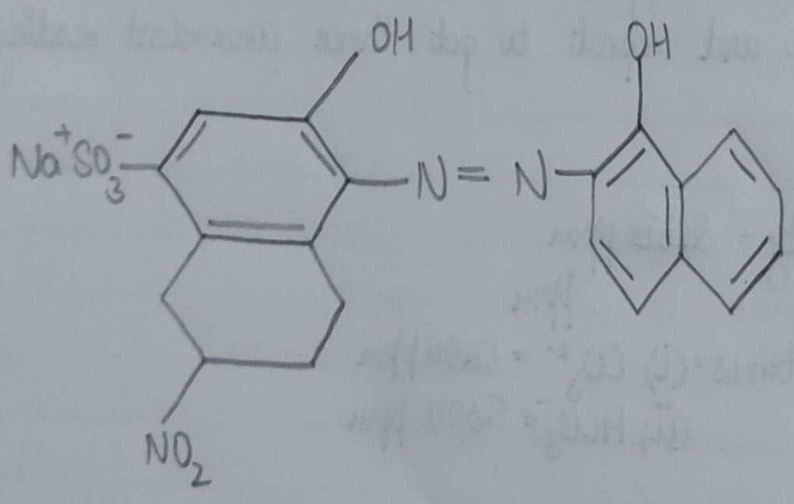
EDTA + metal indicator complex → Metal ion - EDTA + Indicator
[Wine red colour] [Blue]

The various EDTA species ~~are~~ obtained are abbreviated as H_3Y, H_3Y^-, H_3Y^{2-}, H_3Y^{-3} and H_3Y^{4-}. The relative amounts of these. Species is a function of pH.

Structure of EDTA



Structure of Erichrome Black T



At, pH > 12, EDTA exists as Y^{4-}
pH ≤ 8, EDTA exists as HY^{3-}
pH ≈ 5, EDTA exists as H_2Y^{2-}

The reaction takes place at pH = 10 and the buffer is made by ammonium chloride and ammonium solutions.

At, pH < 6, EDTA exists as H_2In^- (Red)
pH < 7, EDTA exists as HIn^{2-} (Blue)
pH < 10, EDTA exists as In^{3-} (orange)

Procedure :-

1. In 250 ml conical flask pipette out 5 ml of prepared calcium solution.
2. Add 10ml of buffer solution, 1-2 drops of EBT indicator to 10ml of distilled water.
3. Add EDTA from burette into the flask till red colour solution change to blue colour.
4. Repeat the titration to get 3 concordent readings.

Result:- The calcium hardness is of the given water sample =

Observation :-

SNo	Burette Reading		Volume of EDTA used [mL]	Concordent Volume [mL]
	Initial	Final		

Calculations -

EXPERIMENT- 4

Aim:- To determine the amount of oxalic acid and sulphuric acid in one litre of solution with given standard NaOH solution and standard $KMnO_4$ solution.

Apparatus Required:- Beaker, pipette, conical flask, burette, hot plate.

Chemicals Required:- Solution of oxalic acid and sulphuric acid, sodium hydroxide (0.1N), $KMnO_4$ solution (0.1N), Phenolphthalein indicator.

Principle:-

This experiment involves double titration.

1st titration

NaOH reacts with oxalic acid as well as H_2SO_4 according to the following equation:-

$$\underset{\text{(sodium hydroxide)}}{2NaOH} + \underset{\text{(oxalic acid)}}{(COOH)_2 \cdot 2H_2O} \longrightarrow \underset{\text{(sodium oxalate)}}{(COONa)_2} + \underset{\text{(water)}}{4H_2O}$$

$$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$$

By titrating NaOH against the given mixture of oxalic acid and sulphuric acid, the total normality of oxalic acid and sulphuric acid can be determined.

2^{nd} titration

The mixture of solution ~~of~~ is titrated with N/10 $KMnO_4$, which will react with oxalic acid in the presence of H_2SO_4.

Teacher's Signature ____________

Observations :-

Table 1 : NaOH vs Mixture

SNo	Burette Readings		Volume of standard NaOH solution used (Final- Initial Reading) (ml)
	Initial Reading	Final Reading	
1			
2			
3			

Concordent volume :-

Table 2 : $KMnO_4$ vs Mixture

SNo	Burette Readings		Volume of standard NaOH solution used [Final- Initial Reading] (ml)
	Initial Reading	Final Reading	
1			
2			
3			
4			

Concordent volume :-

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$
$$(COOH)_2 + (O) \xrightarrow{60\text{-}70^\circ C} 2CO_2 + H_2O$$
$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2.$$

So, the normality of oxalic acid can be find out by 2nd titration and hence, the strength is determined from the normality of the solution obtained in the 1st titration, the normality of oxalic acid is subtracted and, hence the normality of H_2SO_4 and its strength can be find out.

Procedure:-

1st titration

1. Take 0.1N NaOH solution in the washed and rinsed burette and pipette out 10ml of mixture in the titration flask.
2. Add a drop of phenolphthalein and titrate it till the end point.
3. Note down the volume of NaOH used.
4. Repeat these steps to get concordent readings.

2nd titration

1. Wash and rinse the apparatus thoroughly. Take 0.1N $KMnO_4$ solution in burette. Since, it is a coloured solution, we note the upper meniscus for taking the initial and subsequent readings.
2. Pipette out 10ml of the mixture in the titration flask. Add almost 10ml of sulphuric acid and then heat till, the titration flask is unbearable to touch.
3. Titrate till the end point and note the volume of $KMnO_4$ used.
4. Repeat the process to get three concordent readings.

Result:-

1. Strength of oxalic acid → 7.56 g/L
2. Strength of sulphuric acid → 7.35 g/L.

Teacher's Signature

CALCULATIONS :-

from titration-1

$N_T =$

$N_T =$

$N_T =$

from titration 2

$N_o =$

$N_o =$

$N_o =$

where normality of :-

$(H_2SO_4$ + oxalic acid$) =$

oxalic acid $=$

$H_2SO_4 = N_S$

$N_S = N_T - N_o$

Strength of oxalic acid $= N_o \times 63$ g/L

Strength of sulphuric acid $= N_S \times 49$ g/L

EXPERIMENT-5

Aim:- To find out the viscosity of given liquid using Ostwald's viscometer.

Requirements:- Ostwald's ~~process~~ viscometer, stopwatch, specific gravity bottle, pipette, distilled water, rubber tubing

Principle:- The ostwald's viscometer method is based on Poiseville's equation. This relates the rate of flow of a liquid through a capillary tube with the ~~coeff~~ coefficient of viscosity and is expressed by following equation:-

$$\eta = \frac{4\pi r t P}{8 v L}$$

where, v = volume of the liquid of viscosity flowing in time 't', through capillary tube of radius 'r' and length 'L'.

P = Hydrostatic pressure of the liquid.

Thus, the determination of the absolute viscosity by means by Poiseville's expression involves the determination of v, r, t, L, P.

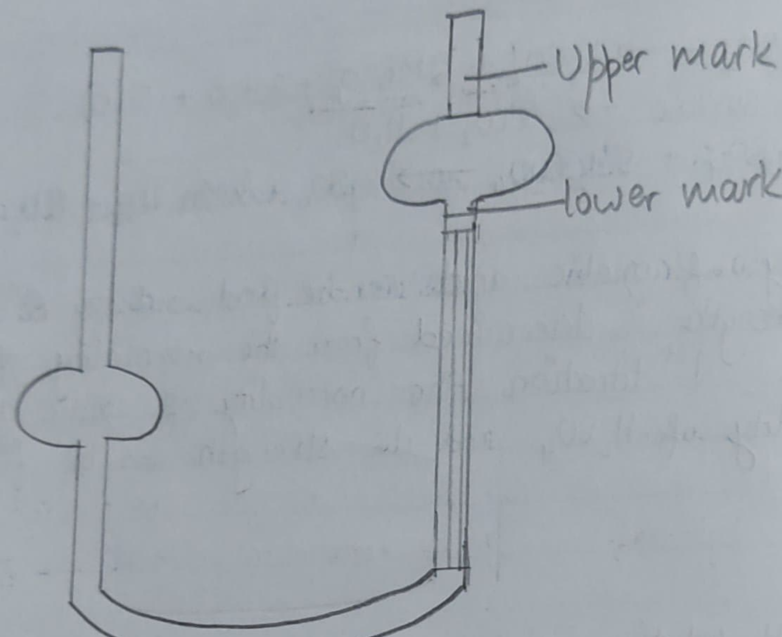
The method is however, tedious and laborious one. Hence, a simpler method is used where in we compare the viscosities of the 2 liquids. If the coefficient of viscosity of one liquid is known, then that of the other can be calculated. If t_1 and t_2 are flow times required to flow for equal volumes of 2 liquids through the same length of a capillary tube, then from equation 1, we have:-

$$\eta_1 = \frac{4\pi r t_1 P_1}{8 v L}, \quad \eta_2 = \frac{4\pi r t_2 P_2}{8 v L}$$

$$\frac{\eta_1}{\eta_2} = \frac{t_1 P_1}{t_2 P_2} \quad -(2)$$

$$P = hdg$$

where, h = height of the liquid column (constant for all the liquids if



Viscometer

taken for identical points and liquids are taken in equal volumes for a particular sets of observations).

g = acceleration due to gravity

η = viscosity of the liquid.

Since, in this case for 2 liquids h and g are same, hence;

$$\frac{\eta_1}{\eta_2} = \frac{t_1 d_1}{t_2 d_2}$$

This will give us the relative viscosity of the given liquid. Relative velocity has no units. The absolute viscosity is given by:-

$$\eta = \frac{d_1}{d_2} \times \frac{t_1}{t_2} \times \eta_2$$

Procedure:-

1. Wash the viscometer with chromic acid ($K_2Cr_2O_7$ + conc. H_2SO_4) and then rinse it 2-3 times with distilled water. It is finally washed with alcohol and ether and then dried.
2. Attach a piece of clean rubber tube to the end of viscometer and clamp the viscometer vertically in air.
3. Now, introduce a sufficient volume of the given liquid with the help of a pipette in bulb so that the bend portion of the U-tube and more than half of bulb is filled up.
4. Through the rubber tube, suck up the liquid until it rises above the mark. Start the stopwatch. Make sure that there is no air bubble inside the liquid.
5. Now, allow the liquid to fall freely through the capillary upto the mark. Start the stopwatch, and note time t_1 for the flow of liquid from mark X to mark Y.
6. Repeat the experiment thrice the values should be concordant.
7. Remove the liquid and clean and dry the viscometer again.
8. Repeat the experiment by taking the some volume of the distilled water

Observations:-

weight of empty specific gravity bottle = 13.40g

weight of empty specific gravity bottle + distilled water = 37.88g

weight of empty specific gravity + given liquid = 38.09 g

SNo	Time of flow of water sample	Time of flow of solution
1	56.55	58.40
2	55.43	59.20

Calculations :-

Mean of water sample $= \frac{56.55 + 55.43}{2} = \boxed{55.99}$

Mean of solution $= \frac{58.4 + 59.2}{2} = \boxed{58.5}$

Density of solution/water $= d_1/d_2 = \frac{w_2 - w_1}{w_3 - w_2}$

$d_1 = w_2 - w_1 = 37.88 - 13.4 \Rightarrow 24.58$

$d_2 = w_3 - w_2 = 38.09 - 13.4 \Rightarrow 24.69$

$\frac{\eta_1}{\eta_2} = \frac{d_1}{d_2} \times \frac{t_1}{t_2} \Rightarrow \frac{24.58}{24.69} \times \frac{55.99}{58.5} \Rightarrow \frac{1376.23}{1444.36} \Rightarrow \boxed{0.952}$

and note down the time taken for flow of water from mark X to Y. Repeat thrice

(9) Weigh the relative density bottle and note down its weight.

(10) Fill it with the given liquid and weigh it again.

(11) Remove the liquid. Wash it with chromic acid and then distilled water. Dry in the oven. Now fill it with the distilled water and weigh it.

Result:- The relative density of the given solution is 0.952

EXPERIMENT-6

Aim:- To estimate the amount of copper present in the given solution by titrating it with hypo solution.

Apparatus:- Burette, pipette, 2 Beakers, titration flask, stand.

Theory:- This experiment comes under the category of iodometry which is used widely in the analysis of ores, alloys etc.
When excess of KI is added to the solution containing Cu^{2+} ion in neutral/slightly acidic medium quantative liberation of iodine takes place.

$$2\,CuSO_4 \cdot 5H_2O + 4KI \longrightarrow 2CuI_2 + 2K_2SO_4 + 5H_2O$$
$$2\,CuI_2 \rightarrow Cu_2I_2 + I_2$$

This liberation is then titrated against standard $Na_2S_2O_3$ solution using starch solution as indicator near the end point.

$$2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$$
(sodium tetra-
thionate)

Ionic equation will be:-

$$2\,Cu^{2+} + 2I^- \rightarrow Cu^{2+} + I_2 + 2e^-$$
$$I_2 + 2S_2O_3^{2-} + 2e^{2-} \rightarrow S_4O_6^{2-} + 2I^-$$

From the above equation, it is evident:-

$$2Cu^{2+} \equiv I_2 \equiv 2S_2O_3^{2-} \equiv 2e^-$$

The equivalent weight of Cu^{2+} will be one half of the twice, the molecular weight since the reaction involves 2 electrons per 2 moles of Cu^{2+}.

Observations:-

S No.	Burette Readings		Volume of Titration (Final - Initial)
	Initial Reading	Final Reading	
1			
2			
3			

Calculations:-

$$\underset{CuSO_4}{N_1V_1} = \underset{Na_2S_2O_3}{N_2V_2}$$

Strength of $Cu^{2+} = N_1 \times 63.5\ g/L$

Equivalent weight of $Cu^{2+} = \frac{2 \times 63.5}{2} \Rightarrow \boxed{63.5}$

From, the normality equation, normality of Cu^{2+} and hence the strength of Cu^{2+} in hydrated $CuSO_4$ crystals is determined.
The titration fails when any ~~number~~ mineral acid is present in the solution and therefore before commencing the titration, the acid should be neutralised. This is done by dropwise addition of a solution of Na_2CO_3 until a slight precipitate or turbidity appears.

$$2H^+ + CO_3^{2-} \rightarrow CO_2 + H_2O$$

$$Cu^{2+} + CO_3^{2-} \rightarrow \underset{\text{(turbidity)}}{CuCO_3}$$

The turbidity is removed by dropwise addition of dilute CH_3COOH.

$$CuCO_3 + 2CH_3COOH \longrightarrow Cu^{2+} + CO_2 + 2CH_3COO^- + H_2O$$

Moreover, the precipitate of Cu_2I_2 absorbs I_2 from the solution and releases it slowly making the detection of sharp end point difficult so a small amount of NH_4SCN is added ~~to~~ near the end point to displace the absorbed iodine from Cu_2I_2 precipitate.

Indicators :- Starch solution

End point :- Blue to colourless.

Procedure :-

1. Take 10ml of $CuSO_4$ solution in conical flask.
2. Add few drops of sodium carbonate (Na_2CO_3) till bluish white ppt is obtained
3. Dissolve the ppt in dil. acetic acid (CH_3COOH). Add half test tube of KI. ~~Convert~~ Cover the flask and keep it in dark for 2-5 minutes.

(4) Brown coloured solution is obtained
(5) Titrate it against hypo till light yellow colour is ~~obtained~~ obtained.
(6) Now add 1 drop of starch and continue the titration till blue colour changes to white which is the end point.

Result:- Strength of Cu^{2+} :-

Precautions:-
(1) The solution of $Na_2S_2O_3$ is always taken in the burette in iodometric titrations
Sufficient amount of KI solution is to be added for the reason explained earlier.
The indicator should be added just before ~~the~~ the end point.

EXPERIMENT-7

Aim:- Determine the quantity of chloride ions in the given water sample using N/30 $AgNO_3$.

Apparatus:- burette, pipette, 2 beakers, flask, stand.

Procedure:-

1. Fill the burette with silver nitrate solution
2. Pipette out 10ml of chloride solution in a conical flask.
3. Add 1-2 drops of K_2CrO_4 solution
4. Add silver nitrate from the burette shaking the flask constantly a white ppt of AgCl is obtained
5. After addition of small volume of $Ag(NO_3)$, a red colour appears in the flask but disappears upon shaking.
6. Continue the titration till a permanent red brown colour is obtained.
7. Take 3 concordant readings.

Reactions Involved:-

$$Ag(NO_3) + NaCl \longrightarrow NaNO_3 + AgCl$$

$$2AgNO_3 + K_2Cr_2O_4 \longrightarrow Ag_2CrO_4 + 2KNO_3$$

Result:- Strength of Cl^- ions = ~~3.24 g/l~~ ___ g/l

Observations :-

SNo	Burette Readings		Volume of titration (Final - Initial)
	Initial Reading	Final Reading	
1 2 3			

Calculations:- $N_1V_1 = N_2V_2$

$N_1 =$

Strength $= N_1 \times$

Date

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EXPERIMENT-8

Aim:- To determine the relative surface tension at the given sample using drop number method.

Apparatus:- Stalogmometer, beaker, weighing bottle with a lid.

Chemicals Used:- Distilled water, liquid whose surface tension is to be measured.

Theory:- Surface Tension of Liquids:-

The molecules of liquid attract each other by cohensive forces, resulting small distances between the moles. Thus, the compressibility of liquid is lower than that of gas, while the density is higher. These cohensive forces are not strong enough to result into fixed position of molecule. which can be seen in solid matter. Liquid doesnot acquire shape but adopts the shape of contained attractive forces, based on electronic ~~attraction~~ interactions. In bulk of liquid, each molecule is attracted by neighbouring molecules in all directions, hence zero net force. Surface molecules are, in energically unfavourable ~~state~~ which forces liquid to reduce surface area, at fixed volume. Each liquid is characterized by their own surface tension, which decreases with increase in temperature. There is 2 line mark on the stalogmometer, top line above the wide part and bottom line below it. The volume V corresponds to n drops released from stalogmometer, upon decrease of fluid level, from top to bottom. The measurement of surface tension of liquid is based on the fact that the drop of liquid at the lower end of capillary falls down when weight of drop becomes just equal to surface tension.

Teacher's Signature

Observations:-

Weight of empty specific gravity bottle = 13.11 g
Weight of empty specific gravity bottle with water = 37.86 g
Weight of empty specific gravity bottle with liquid = 37.81 g
Weight of empty specific gravity = 24.75 g
Weight of water = 37.86 - 13.11 = 24.75 g
Weight of liquid = 37.81 - 13.11 = 24.70 g

SNo	No. of drops	Droplets of solution
1	54	59
2	52	57
3	50	55

Calculations ⇒

$$n_w = \frac{54+52+50}{3} = 52$$

$$n_l = \frac{59+57+55}{3} = 57$$

Relative surface tension of liquid = $\frac{d_l}{d_w} \times \frac{n_w}{n_l}$

$d_w = 24.75$
$d_l = 24.70$
$n_w = 52$
$n_l = 57$

$$\Rightarrow \frac{24.75}{24.7} \times \frac{52}{57} \Rightarrow \frac{1287}{1407.9} \Rightarrow \boxed{0.914}$$

Date

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Procedure:-

① Fill the stalagmometer upto the top mark with distilled water. Release it to the weighing bottle and count how many drops it takes to decrease the water level in stalagmometer down to bottom mark.

② Empty and dry the weighing bottle and stalogmometer for next measurement.

③ Repeat step ① and ② for liquids with unknown surface tension

④ Write down the densities of studied liquids according to the notes on bottle.

Result:- The relative surface tension of liquid liquids:- 0.914

Teacher's Signature